

Properties of Fourier Transform (Part 1)

1. Linearity: $x_1(t) \Leftrightarrow X_1(j\omega)$
 $x_2(t) \Leftrightarrow X_2(j\omega)$
 $\alpha x_1(t) \Leftrightarrow \alpha X_1(j\omega)$
 $\beta x_2(t) \Leftrightarrow \beta X_2(j\omega)$
 $\alpha x_1(t) + \beta x_2(t) \Leftrightarrow \alpha X_1(j\omega) + \beta X_2(j\omega)$

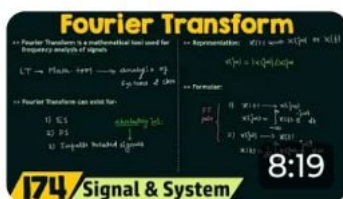
proof:- $x(t) \Leftrightarrow X(j\omega)$
 $X(j\omega) = \int_{-\infty}^{\infty} x(t) \cdot e^{-j\omega t} dt$
 $= \int_{-\infty}^{\infty} [\alpha x_1(t) + \beta x_2(t)] \cdot e^{-j\omega t} dt$
 $= \alpha \int_{-\infty}^{\infty} x_1(t) \cdot e^{-j\omega t} dt + \beta \int_{-\infty}^{\infty} x_2(t) \cdot e^{-j\omega t} dt$

2: Conjugation: $x(t) \Leftrightarrow X(j\omega)$
 $x^*(t) \Leftrightarrow X^*(-j\omega)$

proof:- $x(t) \Leftrightarrow X(j\omega)$
 $X(j\omega) = \int_{-\infty}^{\infty} x(t) \cdot e^{-j\omega t} dt$
 $x^*(t) \Leftrightarrow X^*(j\omega)$
 $X^*(j\omega) = \int_{-\infty}^{\infty} x^*(t) \cdot e^{-j\omega t} dt$
 $\omega \rightarrow -\omega$
 $X^*(j\omega) = \int_{-\infty}^{\infty} x^*(t) \cdot e^{j\omega t} dt$
 $x^*(t) \Leftrightarrow X^*(-j\omega)$

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Introduction to Fourier Transform

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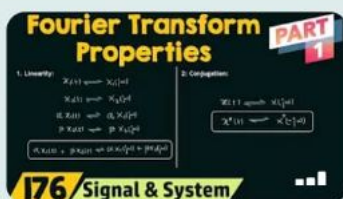
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Conditions for Existence of Fourier Transform (Dirichlet...

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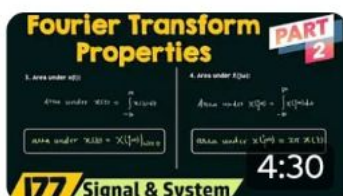
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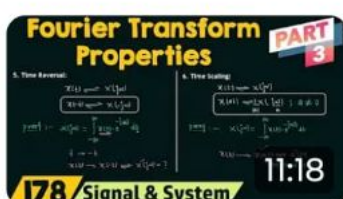
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Properties of Fourier Transform (Part 2)

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Properties of Fourier Transform (Part 3)

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Properties of Fourier

Properties of Fourier Transform (Part 2)

3. Area under $x(t)$:

$$\begin{aligned} \text{Area under } x(t) &= \int_{-\infty}^{\infty} x(t) dt \\ X(j\omega) &= \int_{-\infty}^{\infty} x(t) \cdot e^{-j\omega t} dt \\ &\downarrow \omega=0 \\ X(0) &= \int_{-\infty}^{\infty} x(t) dt = \text{area under } x(t) \\ \text{** area under } x(t) &= X(j\omega)|_{\omega=0} \end{aligned}$$

4. Area under $X(j\omega)$:

$$\begin{aligned} \text{Area under } X(j\omega) &= \int_{-\infty}^{\infty} X(j\omega) d\omega \\ x(t) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) \cdot e^{j\omega t} d\omega \\ &\downarrow t=0 \\ x(0) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) d\omega \\ 2\pi x(0) &= \int_{-\infty}^{\infty} X(j\omega) d\omega = \text{area under } X(j\omega) \\ \text{area under } X(j\omega) &= 2\pi x(0) \end{aligned}$$

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Fourier Transform (Dirichlet Conditions)

1. Signal should have finite number of maxima and minima over any finite interval.
2. Signal should have finite number of discontinuities over any finite interval.
3. Signal should be absolutely integrable.

$$\int_{-\infty}^{\infty} |x(t)| dt < \infty$$

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Conditions for Existence of Fourier Transform (Dirichlet...

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Fourier Transform Properties PART 1

1. Linearity:

$$\begin{aligned} X_1(t) &\longleftrightarrow X_1(j\omega) \\ X_2(t) &\longleftrightarrow X_2(j\omega) \\ \alpha X_1(t) + \beta X_2(t) &\longleftrightarrow \alpha X_1(j\omega) + \beta X_2(j\omega) \end{aligned}$$
2. Superposition:

$$\begin{aligned} X_1(t) &\longleftrightarrow X_1(j\omega) \\ X_2(t) &\longleftrightarrow X_2(j\omega) \\ X_1(t) + X_2(t) &\longleftrightarrow X_1(j\omega) + X_2(j\omega) \end{aligned}$$

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Properties of Fourier Transform (Part 1)

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Fourier Transform Properties PART 2

3. Area under $x(t)$:

$$\text{Area under } x(t) = \int_{-\infty}^{\infty} x(t) dt = X(j\omega)|_{\omega=0}$$
4. Area under $X(j\omega)$:

$$\text{Area under } X(j\omega) = \int_{-\infty}^{\infty} X(j\omega) d\omega = 2\pi x(0)$$

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Fourier Transform Properties PART 3

5. Time Reversal:

$$\begin{aligned} X(t) &\longleftrightarrow X(j\omega) \\ X(-t) &\longleftrightarrow X^*(j\omega) \end{aligned}$$
6. Time Scaling:

$$\begin{aligned} X(t) &\longleftrightarrow X(j\omega) \\ X(at) &\longleftrightarrow \frac{1}{|a|} X\left(\frac{j\omega}{a}\right) \end{aligned}$$

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Properties of Fourier Transform (Part 3)

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Fourier Transform Properties PART 4

7. Time Shifting:

$$\begin{aligned} X(t) &\longleftrightarrow X(j\omega) \\ X(t-t_0) &\longleftrightarrow e^{-j\omega t_0} X(j\omega) \end{aligned}$$
8. Frequency Shifting:

$$\begin{aligned} X(t) &\longleftrightarrow X(j\omega) \\ X(t) e^{j\omega_0 t} &\longleftrightarrow X(j(\omega - \omega_0)) \end{aligned}$$

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Properties of Fourier Transform (Part 4)

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Fourier Transform PART

Properties of Fourier

$t \rightarrow -t$
 $x(t) \rightarrow x(-t) \Rightarrow \underline{x(j\omega) = ?}$

$$\underline{x(j\omega)} = \int_{-\infty}^{\infty} x(-t) \cdot e^{-j\omega t} dt$$

 $-t = \tau \Rightarrow t = -\tau \Rightarrow dt = -d\tau$
 $t = -\infty \Rightarrow \tau = \infty$
 $t = \infty \Rightarrow \tau = -\infty$

$$\underline{x(j\omega)} = \int_{\infty}^{-\infty} x(\tau) \cdot e^{-j\omega(-\tau)} (-d\tau) = \int_{-\infty}^{\infty} x(\tau) \cdot e^{-j\omega \tau} d\tau = x(j\omega) \Rightarrow x(-t)$$

$$\underline{x(j\omega)} = \frac{1}{a} x(j\frac{\omega}{a})$$

$$\underline{x'(j\omega)} = \frac{1}{|a|} x(j\frac{\omega}{a}) ; a \neq 0$$

I) a is positive

$$\underline{x'(j\omega)} = \int_{-\infty}^{\infty} x(at) \cdot e^{-j\omega t} dt$$

 $\frac{at}{a} = \tau \Rightarrow t = \frac{\tau}{a} \Rightarrow dt = \frac{d\tau}{a}$

$$\underline{x'(j\omega)} = \int_{-\infty}^{\infty} x(\tau) \cdot e^{-j\omega \frac{\tau}{a}} \frac{d\tau}{a} = \frac{1}{a} \int_{-\infty}^{\infty} x(\tau) \cdot e^{-j(\frac{\omega}{a})\tau} d\tau = \frac{1}{a} x(j\frac{\omega}{a})$$

 II) a is negative

$$\underline{x'(j\omega)} = \frac{1}{|a|} x(j\frac{\omega}{a})$$

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Fourier Transform Properties PART 1
 1. Linearity

$$x_1(t) \leftrightarrow X_1(j\omega)$$

$$x_2(t) \leftrightarrow X_2(j\omega)$$

$$a x_1(t) + b x_2(t) \leftrightarrow a X_1(j\omega) + b X_2(j\omega)$$

 2. Time Shifting

$$x(t-t_0) \leftrightarrow e^{-j\omega t_0} X(j\omega)$$

 3. Frequency Shifting

$$x(t) e^{j\omega_0 t} \leftrightarrow X(j(\omega - \omega_0))$$

 4. Time Reversal

$$x(-t) \leftrightarrow X^*(j\omega)$$

 5. Area under $x(t)$

$$\int_{-\infty}^{\infty} x(t) dt = X(j0)$$

 6. Area under $X(j\omega)$

$$\int_{-\infty}^{\infty} X(j\omega) d\omega = 2\pi x(0)$$

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Properties of Fourier Transform (Part 1)

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Fourier Transform Properties PART 2
 1. Time Reversal

$$x(-t) \leftrightarrow X^*(j\omega)$$

 2. Area under $x(t)$

$$\int_{-\infty}^{\infty} x(t) dt = X(j0)$$

 3. Area under $X(j\omega)$

$$\int_{-\infty}^{\infty} X(j\omega) d\omega = 2\pi x(0)$$

 4. Time Shifting

$$x(t-t_0) \leftrightarrow e^{-j\omega t_0} X(j\omega)$$

 5. Frequency Shifting

$$x(t) e^{j\omega_0 t} \leftrightarrow X(j(\omega - \omega_0))$$

 6. Linearity

$$a x_1(t) + b x_2(t) \leftrightarrow a X_1(j\omega) + b X_2(j\omega)$$

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Properties of Fourier Transform (Part 2)

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Fourier Transform Properties PART 3
 1. Time Reversal

$$x(-t) \leftrightarrow X^*(j\omega)$$

 2. Area under $x(t)$

$$\int_{-\infty}^{\infty} x(t) dt = X(j0)$$

 3. Area under $X(j\omega)$

$$\int_{-\infty}^{\infty} X(j\omega) d\omega = 2\pi x(0)$$

 4. Time Shifting

$$x(t-t_0) \leftrightarrow e^{-j\omega t_0} X(j\omega)$$

 5. Frequency Shifting

$$x(t) e^{j\omega_0 t} \leftrightarrow X(j(\omega - \omega_0))$$

 6. Linearity

$$a x_1(t) + b x_2(t) \leftrightarrow a X_1(j\omega) + b X_2(j\omega)$$

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Fourier Transform Properties PART 4
 1. Time Reversal

$$x(-t) \leftrightarrow X^*(j\omega)$$

 2. Area under $x(t)$

$$\int_{-\infty}^{\infty} x(t) dt = X(j0)$$

 3. Area under $X(j\omega)$

$$\int_{-\infty}^{\infty} X(j\omega) d\omega = 2\pi x(0)$$

 4. Time Shifting

$$x(t-t_0) \leftrightarrow e^{-j\omega t_0} X(j\omega)$$

 5. Frequency Shifting

$$x(t) e^{j\omega_0 t} \leftrightarrow X(j(\omega - \omega_0))$$

 6. Linearity

$$a x_1(t) + b x_2(t) \leftrightarrow a X_1(j\omega) + b X_2(j\omega)$$

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Properties of Fourier Transform (Part 4)

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Fourier Transform Properties PART 5
 1. Time Reversal

$$x(-t) \leftrightarrow X^*(j\omega)$$

 2. Area under $x(t)$

$$\int_{-\infty}^{\infty} x(t) dt = X(j0)$$

 3. Area under $X(j\omega)$

$$\int_{-\infty}^{\infty} X(j\omega) d\omega = 2\pi x(0)$$

 4. Time Shifting

$$x(t-t_0) \leftrightarrow e^{-j\omega t_0} X(j\omega)$$

 5. Frequency Shifting

$$x(t) e^{j\omega_0 t} \leftrightarrow X(j(\omega - \omega_0))$$

 6. Linearity

$$a x_1(t) + b x_2(t) \leftrightarrow a X_1(j\omega) + b X_2(j\omega)$$

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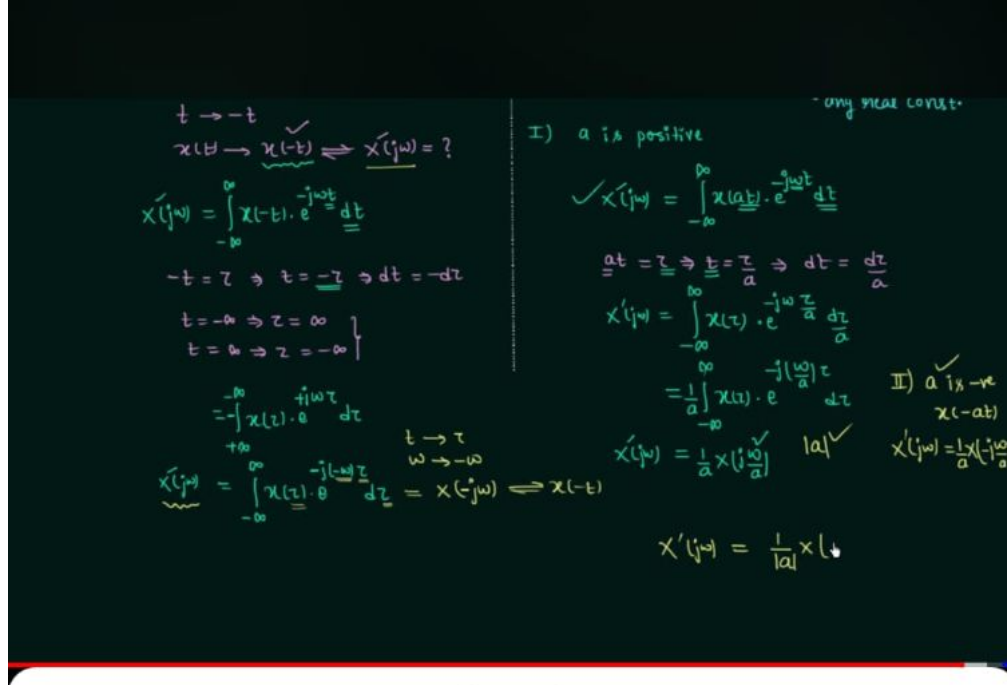
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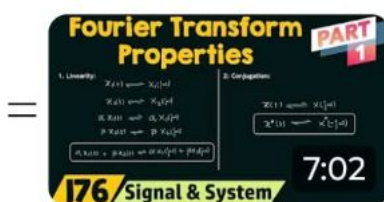
Fourier Transform PART

Properties of Fourier



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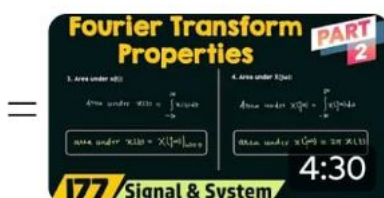
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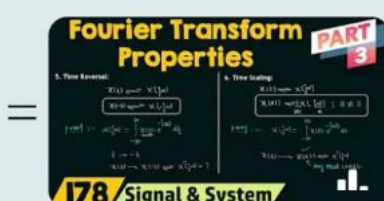
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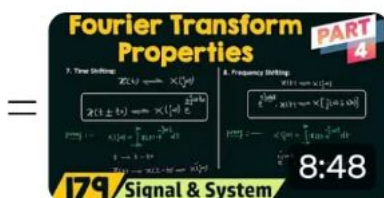
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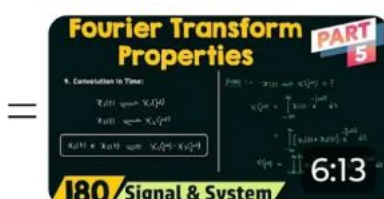
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Properties of Fourier Transform (Part 5)

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Properties of Fourier



Properties of Fourier Transform (Part 3)

5. Time Reversal:

$$\begin{aligned}
 x(t) &\Longleftrightarrow X(j\omega) \\
 x(-t) &\Longleftrightarrow X(-j\omega) \\
 \text{proof: } X(j\omega) &= \int_{-\infty}^{\infty} x(t) \cdot e^{-j\omega t} dt \\
 t &\rightarrow -t \\
 x(t) &\rightarrow x(-t) \Rightarrow X(j\omega) = ? \\
 X(j\omega) &= \int_{-\infty}^{\infty} x(-t) \cdot e^{-j\omega t} dt \\
 -t = \tau &\Rightarrow t = -\tau \Rightarrow dt = -d\tau \\
 t = -\infty &\Rightarrow \tau = \infty \\
 t = \infty &\Rightarrow \tau = -\infty
 \end{aligned}$$

6. Time Scaling:

$$\begin{aligned}
 x(t) &\Longleftrightarrow X(j\omega) \\
 x(at) &\Longleftrightarrow \frac{1}{|a|} X\left(\frac{j\omega}{a}\right); a \neq 0 \\
 \text{proof: } X(j\omega) &= \int_{-\infty}^{\infty} x(t) \cdot e^{-j\omega t} dt \\
 x(t) &\rightarrow x(at) \Rightarrow X(j\omega) = ? \\
 \text{I) } a &\text{ is positive} \\
 X(j\omega) &= \int_{-\infty}^{\infty} x(at) \cdot e^{-j\omega t} dt \\
 at = \tau &\Rightarrow t = \frac{\tau}{a} \Rightarrow dt = \frac{d\tau}{a}
 \end{aligned}$$

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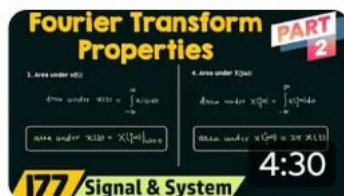
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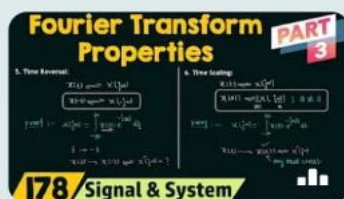
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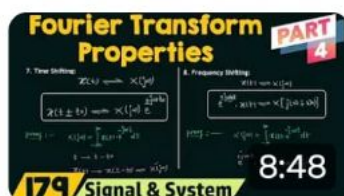
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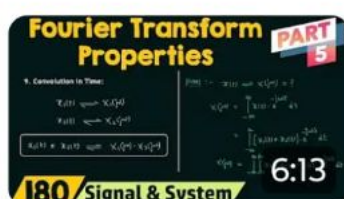
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Properties of Fourier Transform (Part 5)

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Properties of Fourier Transform (Part 6)

Properties of Fourier Transform (Part 4)

7. Time Shifting: $x(t) \Rightarrow X(j\omega)$
 $x(t \pm t_0) \Rightarrow X(j\omega) e^{\pm j\omega t_0}$

Proof: $\checkmark$ $X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$
 $t \rightarrow t - t_0$
 $x(t) \rightarrow x(t - t_0) \Rightarrow X(j\omega)$
 $X(j\omega) = \int_{-\infty}^{\infty} x(t - t_0) e^{-j\omega t} dt$
 $t - t_0 = \tau \Rightarrow t = \tau + t_0 \Rightarrow dt = d\tau$
 $X(j\omega) = \int_{-\infty}^{\infty} x(\tau) e^{-j\omega(\tau + t_0)} d\tau$
 $= \int_{-\infty}^{\infty} x(\tau) e^{-j\omega\tau} e^{-j\omega t_0} d\tau$

8. Frequency Shifting: $x(t) \Rightarrow X(j\omega)$
 $e^{\pm j\omega_0 t} x(t) \Rightarrow X(j(\omega \mp \omega_0))$

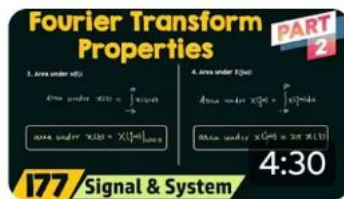
Proof: $\checkmark$ $X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$
 $e^{\pm j\omega_0 t} x(t) \Rightarrow X(j\omega)$
 $X'(j\omega) = \int_{-\infty}^{\infty} e^{\pm j\omega_0 t} x(t) e^{-j\omega t} dt$
 $\checkmark$ $X'(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j(\omega \mp \omega_0)t} dt$
 $X'(j\omega) = X[j(\omega \mp \omega_0)] \Rightarrow e^{\pm j\omega_0 t} x(t)$

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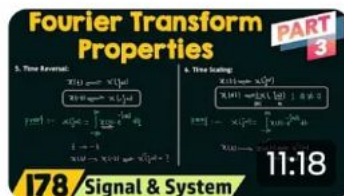
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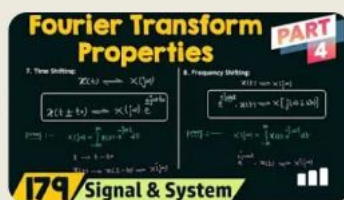
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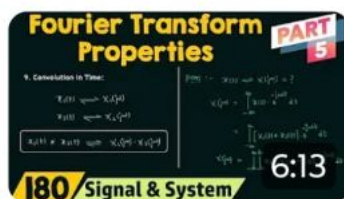
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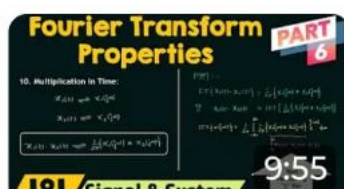
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Properties of Fourier Transform (Part 6)
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$\frac{d}{dt} x(t) \Leftrightarrow j\omega X(\omega)$
Proof :- $x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega$
 $\frac{d}{dt} x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) \cdot \frac{d}{dt} e^{j\omega t} d\omega$
 $\frac{d}{dt} x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) \cdot j\omega e^{j\omega t} d\omega$
 $\frac{d}{dt} x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} (j\omega X(j\omega)) e^{j\omega t} d\omega$
 $\frac{d}{dt} x(t) \Leftrightarrow j\omega X(\omega)$
 $x(t) \cdot \delta(t-t_0) = x(t_0) \cdot \delta(t-t_0)$
Proof :- $x(t) * u(t) = \int_{-\infty}^{\infty} x(\tau) u(t-\tau) d\tau$

 $x(t) * u(t) = \int_{-\infty}^t x(\tau) d\tau$
 $FT[x(t) * u(t)] = FT\{x(t)\} \cdot FT\{u(t)\}$
 $= X(j\omega) \cdot \left(\frac{1}{j\omega} + \pi\delta(\omega)\right)$
 $= \frac{X(j\omega)}{j\omega} + \pi X(j\omega)\delta(\omega)$
 $= \frac{X(j\omega)}{j\omega} + \pi X(0)\delta(\omega)$

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Fourier Transform Properties PART 5
 9. Calculation in Time:
 $x(t) \Leftrightarrow X(\omega)$
 $x_1(t) \Leftrightarrow X_1(\omega)$
 $x_2(t) \Leftrightarrow X_2(\omega)$
 $x_1(t) + x_2(t) \Leftrightarrow X_1(\omega) + X_2(\omega)$
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 6:13

Properties of Fourier Transform (Part 5)

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Fourier Transform Properties PART 6
 10. Multiplication in Time:
 $x(t) \Leftrightarrow X(\omega)$
 $x_1(t) \Leftrightarrow X_1(\omega)$
 $x_2(t) \Leftrightarrow X_2(\omega)$
 $x_1(t) x_2(t) \Leftrightarrow \frac{1}{2\pi} \int_{-\infty}^{\infty} X_1(\omega) X_2(\omega) e^{j\omega t} d\omega$
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 9:55

Properties of Fourier Transform (Part 6)

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Fourier Transform Properties PART 7
 11. Differentiation in Time:
 $x(t) \Leftrightarrow X(\omega)$
 $\frac{d}{dt} x(t) \Leftrightarrow j\omega X(\omega)$
 12. Integration in Time:
 $x(t) \Leftrightarrow X(\omega)$
 $\int_{-\infty}^t x(\tau) d\tau \Leftrightarrow \frac{X(\omega)}{j\omega} + \pi X(0)\delta(\omega)$
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 11:11

Properties of Fourier Transf...

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Fourier Transform Properties PART 8
 13. Differentiation in Frequency:
 $x(t) \Leftrightarrow X(\omega)$
 $t x(t) \Leftrightarrow j \frac{d}{d\omega} X(\omega)$
 14. Modulation:
 $x(t) \Leftrightarrow X(\omega)$
 $x(t) e^{j\omega_0 t} \Leftrightarrow X(\omega - \omega_0)$
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Properties of Fourier Transform (Part 8)

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Parseval's Energy Theorem
 15. Parseval's Energy Theorem:
 $x(t) \Leftrightarrow X(\omega)$
 $E_{x(t)} = \frac{1}{2\pi} \int_{-\infty}^{\infty} |X(\omega)|^2 d\omega$
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Parseval's Energy Theorem

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Fourier Transform

Duality Property of Fourier

Properties of Fourier Transform (Part 7)

11. Differentiation in Time:

$$x(t) \Rightarrow X(j\omega)$$

$$\frac{d}{dt} x(t) \Rightarrow (j\omega) X(j\omega)$$

Proof :-

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega$$

$$\frac{d}{dt} x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) \cdot \frac{d}{dt} e^{j\omega t} d\omega$$

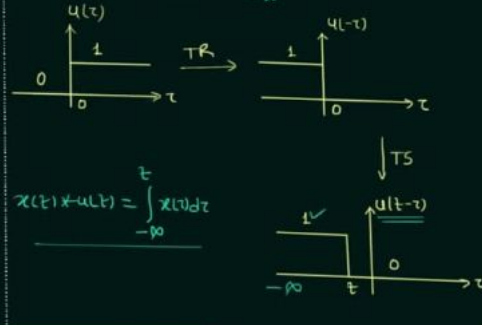
$$\frac{d}{dt} x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) \cdot j\omega e^{j\omega t} d\omega$$

$$\frac{d}{dt} x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} (j\omega) X(j\omega) e^{j\omega t} d\omega$$

12: Integration in Time: $x(t) \Rightarrow X(j\omega)$

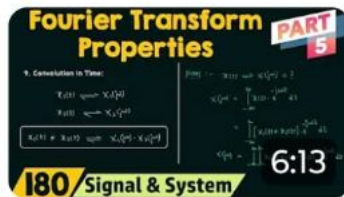
$$\int_{-\infty}^t x(\tau) d\tau \Rightarrow \frac{X(j\omega)}{j\omega} + \pi X(0) \delta(\omega)$$

Proof :- $x(t) * u(t) = \int_{-\infty}^{\infty} x(\tau) u(t-\tau) d\tau$



Fourier Transform | Chapte... · 9/49

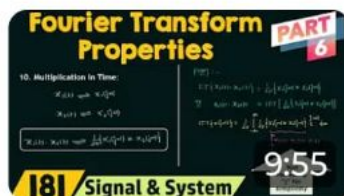
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Properties of Fourier Transform (Part 5)

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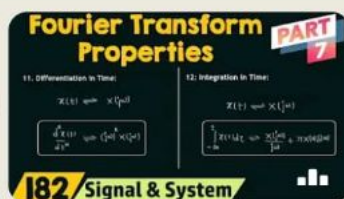
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Properties of Fourier Transform (Part 6)

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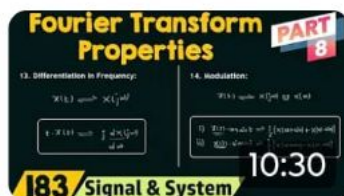
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Properties of Fourier Transf...

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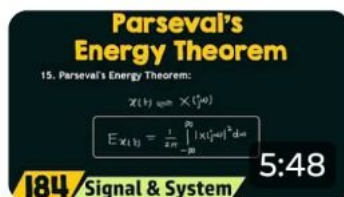
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Properties of Fourier Transform (Part 8)

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1.7 lakh views · 6 years ago



Parseval's Energy Theorem

Neso Academy

2.1 lakh views · 6 years ago



Duality Property of Fourier

Properties of Fourier Transform (Part 8)

13. Differentiation in Frequency:

$$x(t) \Leftrightarrow X(j\omega)$$

$$t \cdot x(t) \Leftrightarrow (j) \frac{d}{d\omega} X(j\omega) \quad n=1$$

Proof: - $X(j\omega) = \int_{-\infty}^{\infty} x(t) \cdot e^{-j\omega t} dt$

$$\frac{dX(j\omega)}{d\omega} = \int_{-\infty}^{\infty} x(t) \cdot \frac{d}{d\omega} e^{-j\omega t} dt$$

$$\frac{dX(j\omega)}{d\omega} = \int_{-\infty}^{\infty} x(t) \cdot (-jt) \cdot e^{-j\omega t} dt$$

$$j \frac{dX(j\omega)}{d\omega} = \int_{-\infty}^{\infty} t \cdot x(t) \cdot e^{-j\omega t} dt$$

FT $\xleftarrow{\text{signal}}$

14. Modulation: $x(t) \Leftrightarrow X(j\omega) \text{ or } X(\omega)$

$$i) x(t) \cdot \cos \omega_0 t \Leftrightarrow \frac{1}{2} [X(\omega + \omega_0) + X(\omega - \omega_0)]$$

$$ii) x(t) \cdot \sin \omega_0 t \Leftrightarrow \frac{j}{2} [X(\omega + \omega_0) - X(\omega - \omega_0)]$$

Proof: - i) $\cos \omega_0 t = \frac{1}{2} [e^{j\omega_0 t} + e^{-j\omega_0 t}]$

$$x(t) \cdot \cos \omega_0 t = \frac{1}{2} [x(t) \cdot e^{j\omega_0 t} + x(t) \cdot e^{-j\omega_0 t}]$$

$$FT\{x(t) \cdot \cos \omega_0 t\} = \frac{1}{2} [FT\{x(t) \cdot e^{j\omega_0 t}\} + FT\{x(t) \cdot e^{-j\omega_0 t}\}]$$

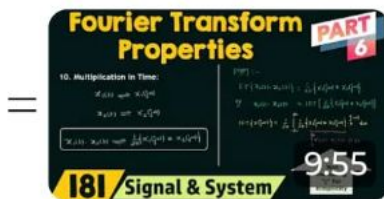
$$\rightarrow X(\omega - \omega_0) + X(\omega + \omega_0)$$

$$= \frac{1}{2} [X(\omega - \omega_0) + X(\omega + \omega_0)]$$

$$ii) \sin \omega_0 t = \frac{j}{2} [e^{j\omega_0 t} - e^{-j\omega_0 t}]$$

Fourier Transform | Chapte... · 10/49

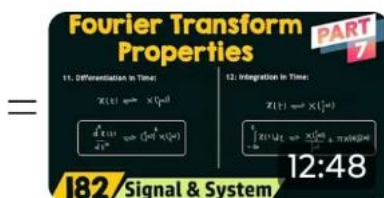
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Properties of Fourier Transform (Part 6)

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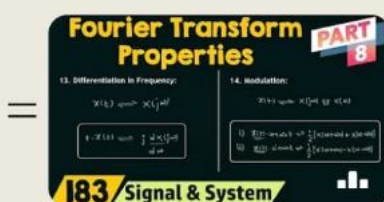
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Properties of Fourier Transform (Part 7)

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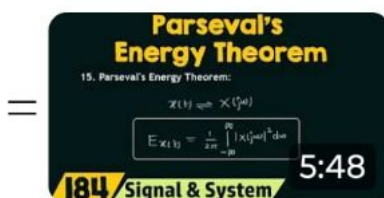
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Properties of Fourier Transf...

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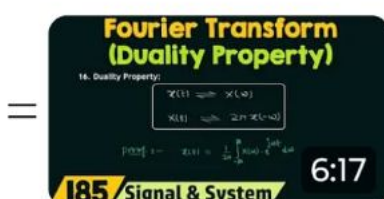
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Parseval's Energy Theorem

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Duality Property of Fourier Transform

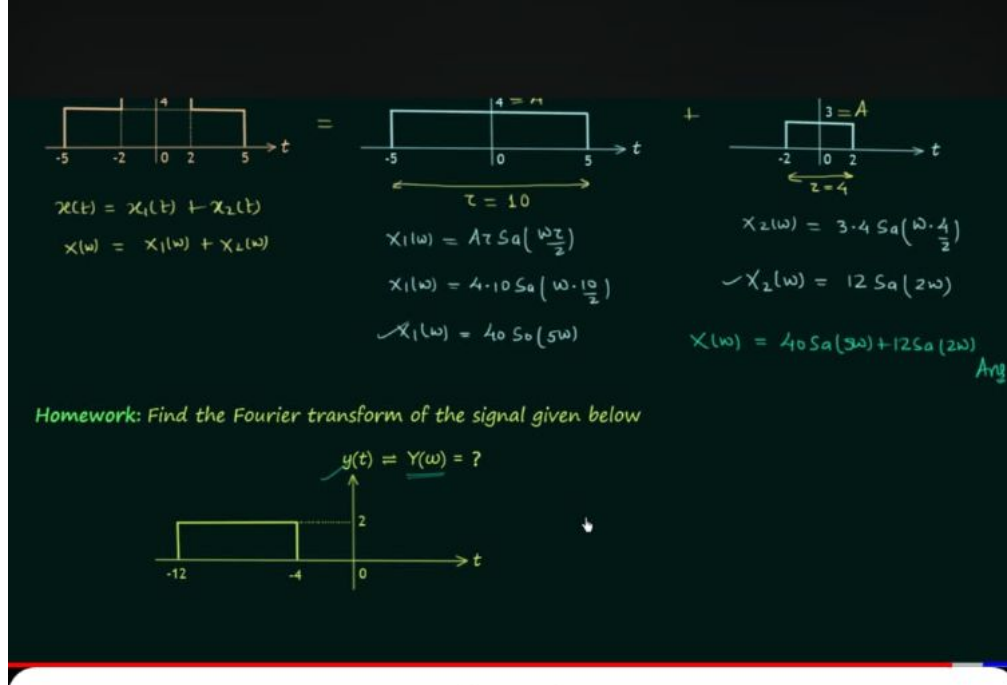
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Fourier Transform of Basic





Fourier Transform | Chapt... · 29/49

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Fourier Transform (Solved Problem 3)

Question: $x(t) = \frac{1}{a + jt} \Rightarrow X(w) = ?$

Homework: $y(t) = \frac{2a}{a^2 + t^2} \Rightarrow Y(w) = ?$

200/Signal & System 2:45

Fourier Transform (Solved Problem 3)

Neso Academy

1.6 lakh views · 6 years ago

Fourier Transform (Solved Problem 4)

Question: Find the Fourier transform of the signal shown below

Homework: 6:59

201/Signal & System

Fourier Transform (Solved Problem 4)

Neso Academy

1.1 lakh views · 6 years ago

Fourier Transform (Solved Problem 5)

Question: Find the Fourier Transform of the signal given below

Homework: 202/Signal & System

Fourier Transform (Solved P...

Neso Academy

1 lakh views · 6 years ago



Fourier Transform (Solved Problem 6)

Question: Let $x(t) = \cos(t - 1/2)$ [where $\cos(t) = 1$ for $-1/2 \leq t \leq 1/2$ and zero elsewhere]. Then if $\cos(t) = \cos(2t)$, the Fourier Transform of $x(t) = x(t - 1)$ will be given by

(a) $\sin(w/2)$
 (b) $\sin(w/2)$
 (c) $\sin(w/2)$
 (d) $\sin(w/2)$

GATE 2008 8:57

203/Signal & System

Fourier Transform (Solved Problem 6)

Neso Academy

85K views · 6 years ago

Fourier Transform (Solved Problem 7)

Question: $x(t) = x_1(t) + x_2(t) \Rightarrow X(w) = ?$

$x_1(t) = \begin{cases} 1 & -1 \leq t \leq 1 \\ 0 & \text{elsewhere} \end{cases}$ $x_2(t) = \begin{cases} 1 & -1 \leq t \leq 1 \\ 0 & \text{elsewhere} \end{cases}$

Homework: 4:03

204/Signal & System

Fourier Transform (Solved Problem 7)

Neso Academy

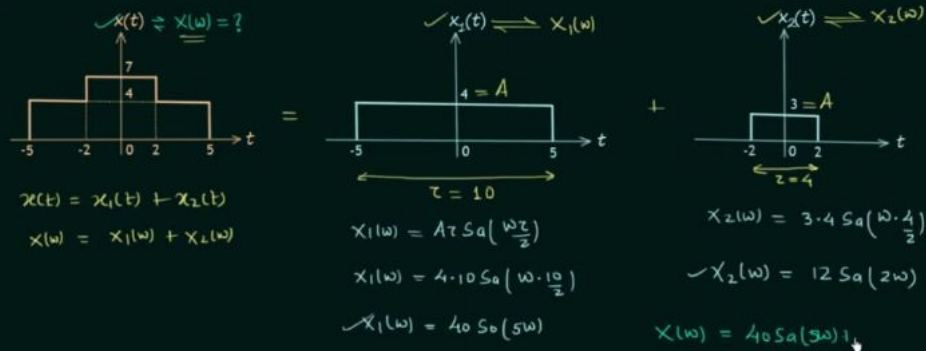
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Fourier Transform

Fourier Transform (Solved

Fourier Transform (Solved Problem 5)

Question: Find the Fourier Transform of the signal given below



Fourier Transform | Chapt... · 29/49

Neso Academy



Fourier Transform (Solved Problem 3)

Question: $x(t) = \frac{1}{a + jt} \Rightarrow X(\omega) = ?$

Homework: $y(t) = \frac{2a}{a^2 + t^2} \Rightarrow Y(\omega) = ?$

200/Signal & System 2:45

Fourier Transform (Solved Problem 3)

Neso Academy

1.6 lakh views · 6 years ago

Fourier Transform (Solved Problem 4)

Question: Find the Fourier transform of the signal shown below

Homework: 6:59

201/Signal & System

Fourier Transform (Solved Problem 4)

Neso Academy

1.1 lakh views · 6 years ago

Fourier Transform (Solved Problem 5)

Question: Find the Fourier Transform of the signal given below

Homework: 202/Signal & System

Fourier Transform (Solved P...

Neso Academy

1 lakh views · 6 years ago



Fourier Transform (Solved Problem 6)

Question: Let $x(t) = \cos(t - 1/2)$ [where $\cos(t) = 1$ for $-1/2 \leq t \leq 1/2$ and zero elsewhere]. Then if $\text{Fourier Transform of } x(t) = X(\omega)$ will be given by

(a) $\sin(\omega/2)$
(b) $\cos(\omega/2)$
(c) $\sin(\omega/2)\cos(\omega/2)$
(d) $\cos(\omega/2)\sin(\omega/2)$

GATE 2008 8:57

203/Signal & System

Fourier Transform (Solved Problem 6)

Neso Academy

85K views · 6 years ago

Fourier Transform (Solved Problem 7)

Question: Let $x(t) = x_1(t) + x_2(t)$ where $x_1(t) = \begin{cases} 1 & -1 \leq t \leq 1 \\ 0 & \text{elsewhere} \end{cases}$ and $x_2(t) = \begin{cases} 1 & 0 \leq t \leq 2 \\ 0 & \text{elsewhere} \end{cases}$. Find the Fourier Transform of $x(t)$.

Homework: 4:03

204/Signal & System

Fourier Transform (Solved Problem 7)

Neso Academy

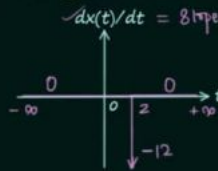
77K views · 6 years ago

Fourier Transform

Fourier Transform (Solved

Fourier Transform (Solved Problem 4)

Question: Find the Fourier transform of the signal shown below



$$\delta(t) \Rightarrow 1$$

$$\frac{d}{dt} x(t) = -12 \delta(t-2)$$

$$(j\omega)^1 x(\omega) = -12 \cdot 1 \cdot e^{-j\omega 2}$$

$$x(\omega) = \frac{-12 e^{-j\omega 2}}{j\omega}$$

$$x(\omega) = \frac{-12 e^{-j\omega 2}}{j\omega} - 4\pi \delta(\omega)$$

$$dc \text{ value} = \frac{4 + (-8)}{2} = -2 \Rightarrow 2\pi \cdot (-2) \cdot \delta(\omega) = -4\pi \delta(\omega)$$

Homework: Find the Fourier transform of the signal shown below



Fourier Transform | Chapt... · 28/49

Neso Academy



Fourier Transform (Solved Problem 2)

Question: If Fourier Transform of $x(t)$ is $X(\omega)$, then find the Fourier Transform of $y(t) = x(2t-5)$

Ans: $Y(\omega) = \frac{1}{2} X\left(\frac{\omega}{2}\right) e^{-j\omega 5/2}$

Homework: $x(t) = \cos(2t)$, find $X(\omega)$

Ans: $X(\omega) = \pi [\delta(\omega - 2) + \delta(\omega + 2)]$

199 Signal & System 6:54

Fourier Transform (Solved Problem 2)

Neso Academy

1.8 lakh views · 6 years ago

Fourier Transform (Solved Problem 3)

Question: $x(t) = \frac{t}{a + jt}$, find $X(\omega)$

Homework: $y(t) = \frac{2a}{a^2 + t^2}$, find $Y(\omega)$

200 Signal & System 2:45

Fourier Transform (Solved Problem 3)

Neso Academy

1.6 lakh views · 6 years ago

Fourier Transform (Solved Problem 4)

Question: Find the Fourier transform of the signal shown below

Ans: $X(\omega) = \frac{1}{\omega} [e^{-j\omega 2} - e^{-j\omega 4}]$

Homework: $x(t) = \cos(2t)$, find $X(\omega)$

Ans: $X(\omega) = \pi [\delta(\omega - 2) + \delta(\omega + 2)]$

201 Signal & System

Fourier Transform (Solved P...

Neso Academy

1.1 lakh views · 6 years ago

Fourier Transform (Solved Problem 5)

Question: Find the Fourier transform of the signal shown below

Ans: $X(\omega) = \frac{1}{\omega} [e^{-j\omega 2} - e^{-j\omega 4}]$

Homework: $x(t) = \cos(2t)$, find $X(\omega)$

Ans: $X(\omega) = \pi [\delta(\omega - 2) + \delta(\omega + 2)]$

202 Signal & System 5:10

Fourier Transform (Solved Problem 5)

Neso Academy

1 lakh views · 6 years ago

Fourier Transform (Solved Problem 6)

Question: Let $x(t) = \cos(2t)$ for $0 \leq t \leq 1$, find $X(\omega)$

Ans: $X(\omega) = \frac{1}{2} [\delta(\omega - 2) + \delta(\omega + 2)]$

Homework: $x(t) = \cos(2t)$, find $X(\omega)$

Ans: $X(\omega) = \pi [\delta(\omega - 2) + \delta(\omega + 2)]$

203 Signal & System 8:57

Fourier Transform (Solved Problem 6)

Neso Academy

85K views · 6 years ago

Fourier Transform

Fourier Transform (Solved

Fourier Transform (Solved Problem 3)

Question: $x(t) = \frac{1}{a + jt} \Rightarrow X(\omega) = ?$

Sol: $\checkmark e^{-at} \cdot u(t), a > 0 \iff \frac{1}{a + j\omega} \quad t = -\omega$

$\frac{1}{a + jt} \xleftrightarrow{\omega = t} \frac{1}{2\pi} \cdot e^{+a\omega} \cdot u(-\omega), a > 0 \iff X(\omega)$

Homework: $y(t) = \frac{2a}{a^2 + t^2} \Rightarrow Y(\omega) = ?$

Fourier Transform | Chapt... · 27/49

Neso Academy



Fourier Transform (Solved Problem 1)

Question: $x(t) = e^{-at^2}, a > 0$
 $X(\omega) = ?$

$\frac{1}{\sqrt{a}} e^{-\frac{\omega^2}{4a}} \iff \int_{-\infty}^{\infty} e^{-\frac{t^2}{4a}} e^{-j\omega t} dt$

198 Signal & System

10:09

Fourier Transform (Solved Problem 1)

Neso Academy

3.8 lakh views · 6 years ago

Fourier Transform (Solved Problem 2)

Question: If Fourier Transform of $x(t)$ is $X(\omega)$, then find the Fourier Transform of $y(t) = x(2t - 3)$

Sol: $\checkmark x(t) \xrightarrow{\text{Time Scaling}} x(2t) \xrightarrow{\text{Time Shifting}} x(2t - 3)$

Homework: $x(t) = e^{-at^2} \Rightarrow X(\omega) = \frac{1}{\sqrt{a}} e^{-\frac{\omega^2}{4a}}$
 $y(t) = x(2t - 3) \Rightarrow Y(\omega) = \frac{1}{2} e^{-j\omega \cdot 3} X(\frac{\omega}{2})$

199 Signal & System

6:54

Fourier Transform (Solved Problem 2)

Neso Academy

1.8 lakh views · 6 years ago

Fourier Transform (Solved Problem 3)

Question: $x(t) = \frac{1}{a + jt} \Rightarrow X(\omega) = ?$

Homework: $y(t) = \frac{2a}{a^2 + t^2} \Rightarrow Y(\omega) = ?$

200 Signal & System

Fourier Transform (Solved P...

Neso Academy

1.6 lakh views · 6 years ago



Fourier Transform (Solved Problem 4)

Question: Find the Fourier transform of the signal shown below

$x(t) = \begin{cases} 1 & -1 \leq t \leq 1 \\ 0 & \text{elsewhere} \end{cases}$

$X(\omega) = \frac{2 \sin \omega}{\omega}$

201 Signal & System

6:59

Fourier Transform (Solved Problem 4)

Neso Academy

1.1 lakh views · 6 years ago

Fourier Transform (Solved Problem 5)

Question: Find the Fourier transform of the signal shown below

$x(t) = \begin{cases} 1 & -1 \leq t \leq 1 \\ 0 & \text{elsewhere} \end{cases}$

$X(\omega) = \frac{2 \sin \omega}{\omega}$

202 Signal & System

5:10

Fourier Transform (Solved Problem 5)

Neso Academy

1 lakh views · 6 years ago

Fourier Transform

Fourier Transform (Solved

Fourier Transform (Solved Problem 2)

Question: If Fourier Transform of $x(t)$ is $X(\omega)$, then find the Fourier Transform of $y(t) = x(2t-3)$

Sol. :- $x(t) \Rightarrow X(\omega)$
 $y(t) = x(2t-3) \Rightarrow Y(\omega) = ?$

M-I :- $x(t) \xrightarrow[\text{Sh.}]{\text{Time}} x(t-3) \xrightarrow[\text{Scaling}]{\text{Time}} x(2t-3)$
 $X(\omega) \longrightarrow X(\omega) \cdot e^{-j\omega 3} \longrightarrow \frac{1}{2} X\left(\frac{\omega}{2}\right) \cdot e^{-j\frac{\omega}{2} 3} \Rightarrow Y(\omega) = \frac{1}{2} X\left(\frac{\omega}{2}\right) \cdot e^{-j\frac{\omega}{2} 3}$

M-II :- $x(t) \xrightarrow[\text{Scaling}]{\text{Time}} x(2t) \xrightarrow[\text{Sh.}]{\text{Time}} x(2t-3) = y(t)$
 $X(\omega) \longrightarrow \frac{1}{2} X\left(\frac{\omega}{2}\right) \longrightarrow \frac{1}{2} X\left(\frac{\omega}{2}\right) \cdot e^{-j\frac{\omega}{2} 3} = Y(\omega)$ Ans

Homework: $x(t) \Rightarrow X(\omega)$
 $y(t) = x(-3t+9) \Rightarrow Y(\omega) = ?$

Fourier Transform | Chapt... · 26/49

Neso Academy



Fourier Transform (Sampling Function)
 $x(t) = \delta(t) \Rightarrow X(\omega) = 1$
 $y(t) = \delta(t) \Rightarrow Y(\omega) = 1$
 $z(t) = \delta(t) \Rightarrow Z(\omega) = 1$
197 Signal & System 6:02

Fourier Transform of Basic Signals (Sampling Function)

Neso Academy

1 lakh views · 6 years ago

Fourier Transform (Solved Problem 1)
 Question: $x(t) = e^{-at^2}$, find $X(\omega)$?
 $x(t) = e^{-at^2}$
198 Signal & System 10:09

Fourier Transform (Solved Problem 1)

Neso Academy

3.8 lakh views · 6 years ago

Fourier Transform (Solved Problem 2)
 Question: If Fourier Transform of $x(t)$ is $X(\omega)$, then find the Fourier Transform of $y(t) = x(2t-3)$
 Sol. :- $x(t) \Rightarrow X(\omega)$
 $y(t) = x(2t-3) \Rightarrow Y(\omega) = ?$
199 Signal & System

Fourier Transform (Solved P...

Neso Academy

1.8 lakh views · 6 years ago



Fourier Transform (Solved Problem 3)
 Question: $x(t) = \frac{1}{a + jt}$, find $X(\omega)$?
 Homework: $y(t) = \frac{2a}{a^2 + t^2}$, find $Y(\omega)$?
200 Signal & System 2:45

Fourier Transform (Solved Problem 3)

Neso Academy

1.6 lakh views · 6 years ago

Fourier Transform (Solved Problem 4)
 Question: Find the Fourier transform of the signal shown below
 $x(t) = \begin{cases} 1 & -1 \leq t \leq 1 \\ 0 & \text{elsewhere} \end{cases}$
 $x(t) = \begin{cases} 1 & -1 \leq t \leq 1 \\ 0 & \text{elsewhere} \end{cases}$
201 Signal & System 6:59

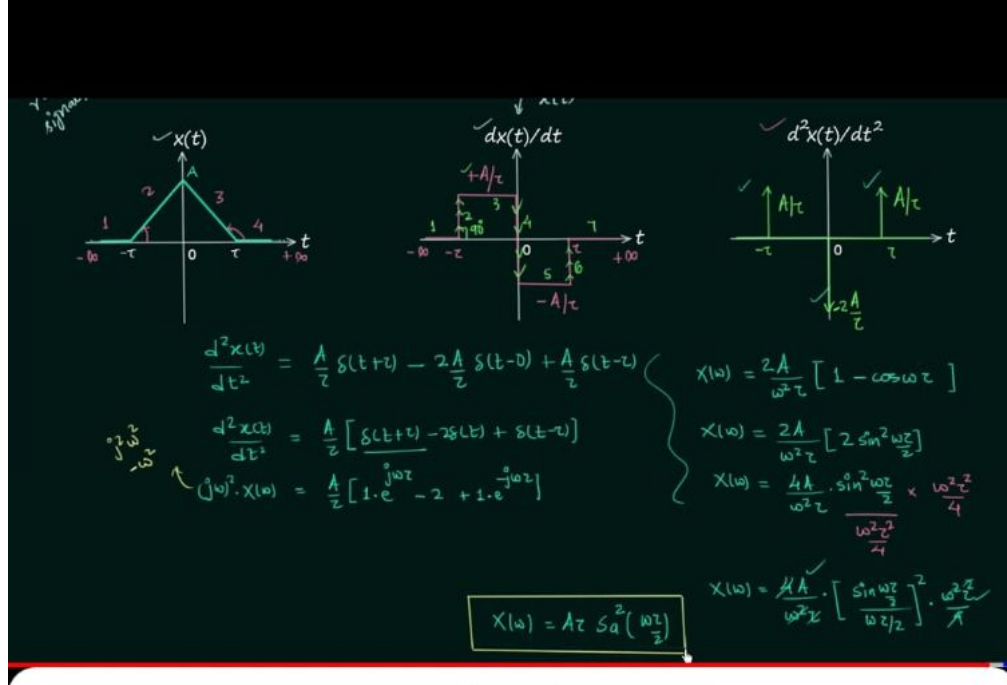
Fourier Transform (Solved Problem 4)

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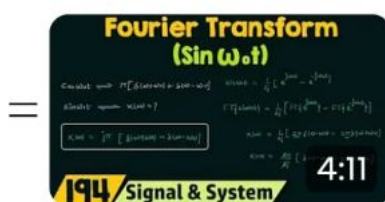
Fourier Transform

Fourier Transform (Solved



Fourier Transform | Chapt... · 23/49

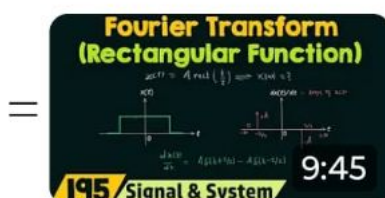
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Fourier Transform of Basic Signals (Sin $\omega_0 t$)

Neso Academy

1.5 lakh views · 6 years ago



Fourier Transform of Basic Signals (Rectangular F...

Neso Academy

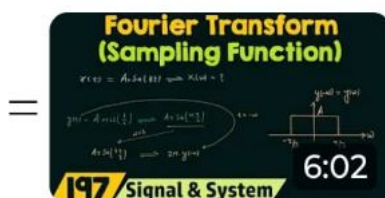
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Fourier Transform of Basic ...

Neso Academy

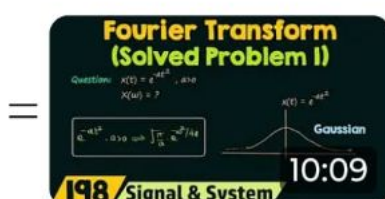
2 lakh views · 6 years ago



Fourier Transform of Basic Signals (Sampling Function)

Neso Academy

1 lakh views · 6 years ago



Fourier Transform (Solved Problem 1)

Neso Academy

3.8 lakh views · 6 years ago



Fourier Transform (Solved Problem 2)

Fourier Transform of Basic Signals (Rectangular Function)

Step signals

$$x(t) = A \text{rect}\left(\frac{t}{\tau}\right) \Rightarrow X(\omega) = ?$$

finite duration signal $\Rightarrow dc = 0$

$$\frac{dx(t)}{dt} = A \delta(t + \tau/2) - A \delta(t - \tau/2)$$

$$\frac{dx(t)}{dt} = A [\delta(t + \tau/2) - \delta(t - \tau/2)]$$

$$(j\omega) X(\omega) = A [1 \cdot e^{j\omega\tau/2} - 1 \cdot e^{-j\omega\tau/2}]$$

$$X(\omega) = \frac{A}{j\omega} [e^{j\omega\tau/2} - e^{-j\omega\tau/2}]$$

$$X(\omega) = \frac{A}{j\omega} \cdot 2j \sin\left(\frac{\omega\tau}{2}\right) = \frac{A\tau}{\omega} \text{Sa}\left(\frac{\omega\tau}{2}\right)$$

area $\times$ duration

Fourier Transform | Chapt... · 22/49

Neso Academy



Fourier Transform (Cos $\omega_0 t$)

193/Signal & System 3:36

Fourier Transform of Basic Signals (Cos $\omega_0 t$)

Neso Academy

2 lakh views · 6 years ago

Fourier Transform (Sin $\omega_0 t$)

194/Signal & System 4:11

Fourier Transform of Basic Signals (Sin $\omega_0 t$)

Neso Academy

1.5 lakh views · 6 years ago

Fourier Transform (Rectangular Function)

195/Signal & System

Fourier Transform of Basic ...

Neso Academy

2.2 lakh views · 6 years ago



Fourier Transform (Triangular Function)

196/Signal & System 11:57

Fourier Transform of Basic Signals (Triangular Function)

Neso Academy

2 lakh views · 6 years ago

Fourier Transform (Sampling Function)

197/Signal & System 6:02

Fourier Transform of Basic Signals (Sampling Function)

Neso Academy

1 lakh views · 6 years ago

Fourier Transform

Fourier Transform (Solved

Fourier Transform of Basic Signals (Sin $\omega_0 t$)

$$\begin{aligned} \checkmark \cos \omega_0 t &\Leftrightarrow \pi [\delta(\omega + \omega_0) + \delta(\omega - \omega_0)] & e^{\pm j\omega_0 t} &\Leftrightarrow 2\pi \delta(\omega \mp \omega_0) \\ \sin \omega_0 t &\Leftrightarrow X(\omega) = ? \\ \sin \omega_0 t &= \frac{1}{2j} [e^{j\omega_0 t} - e^{-j\omega_0 t}] \\ \text{FT}\{\sin \omega_0 t\} &= \frac{1}{2j} [\text{FT}\{e^{j\omega_0 t}\} - \text{FT}\{e^{-j\omega_0 t}\}] \\ X(\omega) &= \frac{1}{2j} [2\pi \delta(\omega - \omega_0) - 2\pi \delta(\omega + \omega_0)] \\ X(\omega) &= \frac{2\pi}{2j} [\delta(\omega - \omega_0) - \delta(\omega + \omega_0)] \\ \checkmark \sin \omega_0 t &\Leftrightarrow X(\omega) = j\pi [\delta(\omega + \omega_0) - \delta(\omega - \omega_0)] \end{aligned}$$

Fourier Transform | Chapte... · 21/49

Neso Academy



Fourier Transform (Complex Exponential Signal)

$x(t) = e^{j\omega_0 t} \Rightarrow X(\omega) = ?$

$X(\omega) = \int_{-\infty}^{\infty} e^{j\omega_0 t} e^{-j\omega t} dt = \int_{-\infty}^{\infty} e^{-j(\omega - \omega_0)t} dt$

$X(\omega) = 2\pi \delta(\omega - \omega_0)$

192/Signal & System

5:34

Fourier Transform of Basic Signals (Complex Exp...

Neso Academy

1.4 lakh views · 6 years ago

Fourier Transform (Cos $\omega_0 t$)

$x(t) = \cos \omega_0 t \Rightarrow X(\omega) = ?$

$\cos \omega_0 t = \frac{1}{2} [e^{j\omega_0 t} + e^{-j\omega_0 t}]$

$X(\omega) = \frac{1}{2} [2\pi \delta(\omega - \omega_0) + 2\pi \delta(\omega + \omega_0)]$

$X(\omega) = \pi [\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$

193/Signal & System

3:36

Fourier Transform of Basic Signals (Cos $\omega_0 t$)

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Fourier Transform (Sin $\omega_0 t$)

$x(t) = \sin \omega_0 t \Rightarrow X(\omega) = ?$

$\sin \omega_0 t = \frac{1}{2j} [e^{j\omega_0 t} - e^{-j\omega_0 t}]$

$X(\omega) = \frac{1}{2j} [2\pi \delta(\omega - \omega_0) - 2\pi \delta(\omega + \omega_0)]$

$X(\omega) = j\pi [\delta(\omega + \omega_0) - \delta(\omega - \omega_0)]$

194/Signal & System

3:36

Fourier Transform of Basic ...

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Fourier Transform (Rectangular Function)

$x(t) = A \text{ rect}\left(\frac{t}{\tau}\right) \Rightarrow X(\omega) = ?$

$X(\omega) = \int_{-\tau/2}^{\tau/2} A e^{-j\omega t} dt = A \tau \text{sinc}\left(\frac{\omega \tau}{2}\right)$

195/Signal & System

9:45

Fourier Transform of Basic Signals (Rectangular F...

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Fourier Transform (Triangular Function)

$x(t) = A \text{ tri}\left(\frac{t}{\tau}\right) \Rightarrow X(\omega) = ?$

$X(\omega) = \int_{-\tau/2}^{\tau/2} A \left(1 - \frac{2|t|}{\tau}\right) e^{-j\omega t} dt = A \tau \text{sinc}^2\left(\frac{\omega \tau}{2}\right)$

196/Signal & System

11:57

Fourier Transform of Basic Signals (Triangular Function)

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Fourier Transform

Fourier Transform of Basic

Fourier Transform of Basic Signals (Cos $\omega_0 t$)

$$\frac{1}{2} e^{j\omega_0 t} \iff 2\pi \delta(\omega - \omega_0) \quad \sin \omega_0 t$$

$$x(t) = \cos \omega_0 t \iff X(\omega) = ?$$

$$\cos \omega_0 t = \frac{1}{2} [e^{j\omega_0 t} + e^{-j\omega_0 t}]$$

$$FT\{\cos \omega_0 t\} = \frac{1}{2} [FT\{e^{j\omega_0 t}\} + FT\{e^{-j\omega_0 t}\}]$$

$$X(\omega) = \frac{1}{2} [2\pi \delta(\omega - \omega_0) + 2\pi \delta(\omega + \omega_0)]$$

$$\cos \omega_0 t \iff \pi [\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$$

$$X(\omega) = \frac{\pi}{2} [\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$$

$$X(\omega) = \pi [\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$$

Fourier Transform | Chapt... · 20/49

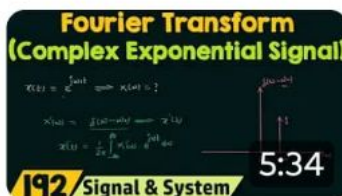
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Fourier Transform of sgn(t) & u(t) [Important Shortcut]

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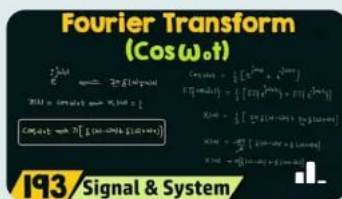
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Fourier Transform of Basic Signals (Complex Exp...)

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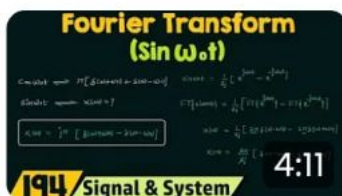
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Fourier Transform of Basic ...

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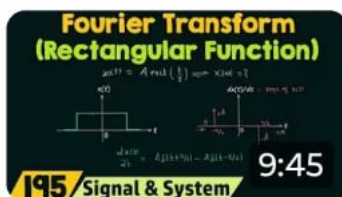
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Fourier Transform of Basic Signals (Sin $\omega_0 t$)

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Fourier Transform of Basic Signals (Rectangular F...)

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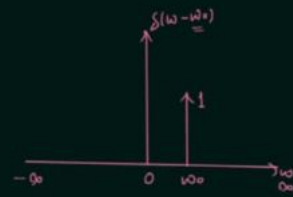
Fourier Transform of Basic

Fourier Transform of Basic Signals ($e^{j\omega_0 t}$)

$$x(t) = e^{j\omega_0 t} \Rightarrow X(\omega) = ?$$

$$\rightarrow X(\omega) = \delta(\omega - \omega_0) \Rightarrow x(t)$$

$$\begin{aligned} x(t) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) \cdot e^{j\omega t} d\omega \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \delta(\omega - \omega_0) \cdot e^{j\omega t} d\omega \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \delta(\omega - \omega_0) \cdot e^{j\omega_0 t} d\omega \\ &= \frac{e^{j\omega_0 t}}{2\pi} \int_{-\infty}^{\infty} \delta(\omega - \omega_0) d\omega \\ X(t) &= \frac{e^{j\omega_0 t}}{2\pi} \cdot 1 \end{aligned}$$



$$2\pi x \frac{e^{j\omega_0 t}}{2\pi} \Rightarrow 2\pi \delta(\omega - \omega_0) \quad \text{for } \omega_0 t \text{ sin } \omega_0 t$$

$$e^{j\omega_0 t} \Rightarrow 2\pi \delta(\omega - \omega_0)$$

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Fourier Transform (Unit Step Signal)

7:05

190/Signal & System

Fourier Transform of Basic Signals (Unit Step Signal)

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Fourier Transform of sgn(t) & u(t) (Shortcut)

12:33

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Fourier Transform of sgn(t) & u(t) [Important Shortcut]

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Fourier Transform (Complex Exponential Signal)

192/Signal & System

Fourier Transform of Basic ...

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Fourier Transform (Cos $\omega_0 t$)

3:36

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Fourier Transform of Basic Signals (Cos $\omega_0 t$)

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Fourier Transform (Sin $\omega_0 t$)

4:11

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Fourier Transform of Basic Signals (Sin $\omega_0 t$)

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Fourier Transform

Fourier Transform of

Fourier Transform of $\text{sgn}(t)$ & $u(t)$ (Shortcut)

Ramp $\rightarrow$ Step $\rightarrow$ Impulse

Signum Function: $x(t) = \text{sgn}(t) \Rightarrow \frac{dx(t)}{dt} = \delta(t)$

Unit Step Signal: $x(t) = u(t) \Rightarrow \frac{dx(t)}{dt} = \delta(t)$

dc value = $\frac{1+(-1)}{2} = 0$

$\frac{dx(t)}{dt} = 2\delta(t)$

$(j\omega)X(\omega) = 2 \cdot 1$

$X(\omega) = \frac{2}{j\omega} + 0$

Unit Step Signal: $x(t) = u(t) \Rightarrow \frac{dx(t)}{dt} = \delta(t)$

dc value = $\frac{1+0}{2} = \frac{1}{2}$

$\frac{dx(t)}{dt} = \delta(t)$

$(j\omega)X(\omega) = 1$

$X(\omega) = \frac{1}{j\omega} + \pi\delta(\omega)$

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Fourier Transform (Signum Function)

$x(t) = \text{sgn}(t) = \begin{cases} 1 & t > 0 \\ 0 & t = 0 \\ -1 & t < 0 \end{cases}$

$\frac{dx(t)}{dt} = \delta(t)$

189 Signal & System 8:03

Fourier Transform of Basic Signals (Signum Function)

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Fourier Transform (Unit Step Signal)

$x(t) = u(t) = \begin{cases} 1 & t > 0 \\ 0 & t < 0 \end{cases}$

$\frac{dx(t)}{dt} = \delta(t)$

190 Signal & System 7:05

Fourier Transform of Basic Signals (Unit Step Signal)

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Fourier Transform of $\text{sgn}(t)$ & $u(t)$ (Shortcut)

Signum Function: $x(t) = \text{sgn}(t)$

Unit Step Signal: $x(t) = u(t)$

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Fourier Transform of $\text{sgn}(t)$...

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Fourier Transform (Complex Exponential Signal)

$x(t) = e^{j\omega_0 t}$

$\frac{dx(t)}{dt} = j\omega_0 e^{j\omega_0 t}$

192 Signal & System 5:34

Fourier Transform of Basic Signals (Complex Exp...)

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Fourier Transform (Cos $\omega_0 t$)

$x(t) = \cos(\omega_0 t)$

$\frac{dx(t)}{dt} = -\omega_0 \sin(\omega_0 t)$

193 Signal & System 3:36

Fourier Transform of Basic Signals (Cos $\omega_0 t$)

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Fourier Transform

Fourier Transform of Basic

Fourier Transform of Basic Signals (Step Signal)

$x(t) = u(t) = \begin{cases} 0, & t < 0 \\ 1, & t > 0 \end{cases} \Rightarrow X(\omega) = ?$

$\text{sgn}(t) \Rightarrow \frac{2}{j\omega}$

$u(t) = \frac{1 + \text{sgn}(t)}{2}$

$\text{FT}\{u(t)\} = \text{FT}\left\{\frac{1}{2}\right\} + \frac{1}{2} \text{FT}\{\text{sgn}(t)\}$

$u(t) = |u(t)| e^{-at} u(t)$

$\int_{-\infty}^{\infty} |u(t)| dt = \infty$

$u(t) = \lim_{a \rightarrow 0} \frac{1}{a} e^{-at} u(t)$

$1 + \text{sgn}(t) = 2u(t)$

$A_0 \Rightarrow 2\pi A_0 \delta(\omega)$

$X(\omega) = 2\pi \times \frac{1}{2} \delta(\omega) + \frac{1}{2} \times \frac{2}{j\omega}$

$X(\omega) = \pi \delta(\omega) + \frac{1}{j\omega}$

$u(t) \Rightarrow \pi \delta(\omega) + \frac{1}{j\omega}$

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Fourier Transform (Exponential Signal)

188 / Signal & System 9:06

Fourier Transform of Basic Signals (Exponential Signals)

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Fourier Transform (Signum Function)

189 / Signal & System 8:03

Fourier Transform of Basic Signals (Signum Function)

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Fourier Transform (Unit Step Signal)

190 / Signal & System

Fourier Transform of Basic ...

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Fourier Transform of sgn(t) & u(t) (Shortcut)

191 / Signal & System 12:33

Fourier Transform of sgn(t) & u(t) [Important Shortcut]

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Fourier Transform (Complex Exponential Signal)

192 / Signal & System 5:34

Fourier Transform of Basic Signals (Complex Exp...)

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Fourier Transform

Fourier Transform of Basic

Fourier Transform of Basic Signals (Signum Function)

$$x(t) = \text{sgn}(t) = \begin{cases} -1, & t < 0 \\ 0, & t = 0 \\ 1, & t > 0 \end{cases} \Rightarrow X(\omega) = ?$$

$$\int_{-\infty}^{\infty} |\text{sgn}(t)| dt < \infty$$

$$u(t) = \lim_{a \rightarrow 0} e^{-at} u(t)$$

$$u(-t) = \lim_{a \rightarrow 0} e^{at} u(-t)$$

$$\text{sgn}(t) = u(t) - u(-t)$$

$$\text{sgn}(t) = \lim_{a \rightarrow 0} [e^{-at} u(t) - e^{at} u(-t)]$$

$$X(\omega) = \lim_{a \rightarrow 0} \left[\frac{1}{a+j\omega} - \frac{1}{a-j\omega} \right]$$

$$= \lim_{a \rightarrow 0} \left[\frac{a-j\omega - a-j\omega}{a^2 - j^2\omega^2} \right]$$

$$= \lim_{a \rightarrow 0} \frac{-2j\omega}{a^2 + \omega^2} \xrightarrow{-1}$$

$$= \frac{-2j\omega}{\omega^2}$$

$$X(\omega) = \frac{-2j}{\omega} \times \frac{j}{j} = \frac{-2j^2}{j\omega}$$

$$X(\omega) = \frac{2}{j\omega}$$

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Fourier Transform (Impulse Signal)

Impulse related signals are absolutely integrable signals and hence we can use the formula of Fourier Transform

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Fourier Transform of Basic Signals (Impulse Signal)

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Fourier Transform (Exponential Signal)

188/Signal & System 9:06

Fourier Transform of Basic Signals (Exponential Signals)

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Fourier Transform (Signum Function)

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Fourier Transform of Basic ...

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Fourier Transform (Unit Step Signal)

190/Signal & System 7:05

Fourier Transform of Basic Signals (Unit Step Signal)

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Fourier Transform of sgn(t) & u(t) (Shortcut)

191/Signal & System 12:33

Fourier Transform of sgn(t) & u(t) [Important Shortcut]

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Fourier Transform

Fourier Transform of

$$X(\omega) = ?$$

$$X(\omega) = \int_{-\infty}^{\infty} \underline{x(t)} \cdot \underline{e^{-j\omega t}} dt$$

$$= \int_{-\infty}^{\infty} \underline{e^{-at}} \cdot \underline{u(t)} \cdot \underline{e^{-j\omega t}} dt$$

$$= \int_0^{\infty} \underline{e^{-at}} \cdot \underline{1} \cdot \underline{e^{-j\omega t}} dt$$

$$= \int_0^{\infty} \underline{e^{-(a+j\omega)t}} dt$$

$$= \left[\frac{e^{-(a+j\omega)t}}{-(a+j\omega)} \right]_0^{\infty}$$

$$= \frac{-1}{a+j\omega} \left[\cancel{e^{-\infty}} - \cancel{e^0} \right]$$

$$X(\omega) = \frac{1}{a+j\omega}$$

$$\rightarrow X(\omega) = \int_{-\infty}^{\infty} \underline{e^{at}} \cdot \underline{u(-t)} \cdot \underline{e^{-j\omega t}} dt$$

$$= \int_{-\infty}^0 \underline{e^{at}} \cdot \underline{1} \cdot \underline{e^{-j\omega t}} dt$$

$$= \int_{-\infty}^0 \underline{e^{(a-j\omega)t}} dt$$

$$= \left[\frac{e^{(a-j\omega)t}}{a-j\omega} \right]_{-\infty}^0$$

$$= \frac{1}{a-j\omega} \left[\cancel{e^0} - \cancel{e^{-\infty}} \right]$$

$$X(\omega) = \frac{1}{a-j\omega}$$

$$e^{at} u(-t) \Rightarrow \frac{1}{a-j\omega} ; a > 0$$

$$\rightarrow X(\omega) = \int_{-\infty}^{\infty} \underline{e^{-a|t|}} \cdot \underline{e^{-j\omega t}} dt$$

$$= \int_{-\infty}^0 \underline{e^{at}} \cdot \underline{e^{-j\omega t}} dt +$$

$$\int_0^{\infty} \underline{e^{-at}} \cdot \underline{e^{-j\omega t}} dt$$

$$= \frac{1}{a-j\omega} + \frac{1}{a+j\omega}$$

$$= \frac{\cancel{a+j\omega} + \cancel{a-j\omega}}{a^2 - j^2\omega^2}$$

$$X(\omega) = \frac{2a}{a^2 + \omega^2} \quad \leftarrow -1$$

$$e^{-a|t|} \Rightarrow \frac{2a}{a^2 + \omega^2} ; a > 0$$

$$X(\omega) = ?$$

$$X(\omega) = \int_{-\infty}^{\infty} \underline{x(t)} \cdot \underline{e^{-j\omega t}} dt$$

$$= \int_{-\infty}^{\infty} \underline{e^{-at}} \cdot \underline{u(t)} \cdot \underline{e^{-j\omega t}} dt$$

$$= \int_0^{\infty} \underline{e^{-at}} \cdot \underline{1} \cdot \underline{e^{-j\omega t}} dt$$

$$= \int_0^{\infty} \underline{e^{-(a+j\omega)t}} dt$$

$$= \left[\frac{e^{-(a+j\omega)t}}{-(a+j\omega)} \right]_0^{\infty}$$

$$= \frac{-1}{a+j\omega} \left[\cancel{e^{-\infty}} - \cancel{e^0} \right]$$

$$X(\omega) = \frac{1}{a+j\omega}$$

$$\rightarrow X(\omega) = \int_{-\infty}^{\infty} \underline{e^{at}} \cdot \underline{u(-t)} \cdot \underline{e^{-j\omega t}} dt$$

$$= \int_{-\infty}^0 \underline{e^{at}} \cdot \underline{1} \cdot \underline{e^{-j\omega t}} dt$$

$$= \int_{-\infty}^0 \underline{e^{(a-j\omega)t}} dt$$

$$= \left[\frac{e^{(a-j\omega)t}}{a-j\omega} \right]_{-\infty}^0$$

$$= \frac{1}{a-j\omega} \left[\cancel{e^0} - \cancel{e^{-\infty}} \right]$$

$$X(\omega) = \frac{1}{a-j\omega}$$

$$e^{at} u(-t) \Rightarrow \frac{1}{a-j\omega} ; a > 0$$

$$\rightarrow X(\omega) = \int_{-\infty}^{\infty} \underline{e^{-a|t|}} \cdot \underline{e^{-j\omega t}} dt$$

$$= \int_{-\infty}^0 \underline{e^{at}} \cdot \underline{e^{-j\omega t}} dt +$$

$$\int_0^{\infty} \underline{e^{-at}} \cdot \underline{e^{-j\omega t}} dt$$

$$= \frac{1}{a-j\omega} + \frac{1}{a+j\omega}$$

$$= \frac{\cancel{a+j\omega} + \cancel{a-j\omega}}{a^2 - j^2\omega^2}$$

$$X(\omega) = \frac{2a}{a^2 + \omega^2} \quad \leftarrow -1$$

$$e^{-a|t|} \Rightarrow \frac{2a}{a^2 + \omega^2} ; a > 0$$

Fourier Transform of Basic Signals (Exponential Signals)

$$i) = \underline{e^{-at} u(t)}, a > 0$$

$$X(\omega) = ?$$

$$X(\omega) = \int_{-\infty}^{\infty} \underline{x(t)} \cdot e^{-j\omega t} dt$$

$$= \int_{-\infty}^{\infty} e^{-at} \underline{u(t)} \cdot e^{-j\omega t} dt$$

$$= \int_0^{\infty} \underline{e^{-at} \cdot 1 \cdot e^{-j\omega t}} dt$$

$$= \int_0^{\infty} e^{-(a+j\omega)t} dt$$

$$= \left[\frac{e^{-(a+j\omega)t}}{-(a+j\omega)} \right]_0^{\infty}$$

$$ii) x(t) = \underline{e^{at} u(-t)}; a > 0$$

$$\rightarrow X(\omega) = \int_{-\infty}^{\infty} e^{at} \underline{u(-t)} \cdot e^{-j\omega t} dt$$

$$= \int_{-\infty}^0 \underline{e^{at} \cdot 1 \cdot e^{-j\omega t}} dt$$

$$= \int_{-\infty}^0 e^{(a-j\omega)t} dt$$

$$= \left[\frac{e^{(a-j\omega)t}}{a-j\omega} \right]_{-\infty}^0$$

$$= \frac{1}{a-j\omega} \left[\underset{1}{e^0} - \underset{0}{e^{-\infty}} \right]$$

$$X(\omega) = \frac{1}{a-j\omega}$$

$$iii) x(t) = \underline{e^{-a|t|}}; a > 0$$

$$\rightarrow X(\omega) = \int_{-\infty}^{\infty} e^{-a|t|} \cdot e^{-j\omega t} dt$$

$$= \int_{-\infty}^0 e^{at} \cdot e^{-j\omega t} dt +$$

$$\int_0^{\infty} e^{-at} \cdot e^{-j\omega t} dt$$

$$= \frac{1}{a-j\omega} + \frac{1}{a+j\omega}$$

$$= \frac{a+j\omega + a-j\omega}{a^2 - j^2\omega^2}$$

$$X(\omega) = \frac{2a}{a^2 + \omega^2}$$

Fourier Transform of Basic Signals (Impulse Signal)

$$x(t) = \delta(t) \iff X(\omega)$$

$$X(\omega) = \int_{-\infty}^{\infty} x(t) \cdot e^{j\omega t} dt$$

Impulse related signals are absolutely integrable signals and hence we can use the formula of Fourier Transform

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Fourier Transform

→ Fourier Transform is a mathematical tool used for frequency analysis of signals.

LT $\Rightarrow f(t) \leftrightarrow F(\omega)$ where ω is angular frequency of signal

→ Fourier Transform can exist for:

1. D.S. continuous signal
2. D.S. discrete signal
3. Single-valued periodic signals

→ Representation: $x(t) = \sum_{-\infty}^{\infty} X(\omega) e^{j\omega t}$

$X(\omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$

→ Fourier:

$$f(t) \leftrightarrow F(\omega) = \int_{-\infty}^{\infty} f(t) e^{-j\omega t} dt$$

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{j\omega t} d\omega$$

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Signal & System

8:19

Introduction to Fourier Transform

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**Fourier Transform
(Dirichlet Conditions)**

1. Signal should have finite number of maxima and minima over any finite interval
2. Signal should have finite number of discontinuities over any finite interval
3. Signal should be absolutely integrable

$$\int_{-\infty}^{\infty} |x(t)| dt < \infty$$

175 Signal & System **8:24**

Conditions for Existence of Fourier Transform (Dirichlet...

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Fourier Transform Properties

1. Linearity

$$X_1(t) \leftrightarrow X_1(f)$$

$$X_2(t) \leftrightarrow X_2(f)$$

$$aX_1(t) + bX_2(t) \leftrightarrow aX_1(f) + bX_2(f)$$

2. Conjugation

$$X_1(t) \leftrightarrow X_1(f)$$

$$X_1^*(t) \leftrightarrow X_1^*(-f)$$

PART 1

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7:02

Properties of Fourier Transform (Part 1)

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Fourier Transform Properties

3. Area under $x(t)$:

$$A_{\text{area under } x(t)} = \int_{-\infty}^{\infty} x(t) dt$$

$A_{\text{area under } X(f)} = X(f=0)$

4. Area under $X(f)$:

$$A_{\text{area under } X(f)} = \int_{-\infty}^{\infty} X(f) df$$

$A_{\text{area under } x(t)} = 2\pi X(f=0)$

PART 2

4:30

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Properties of Fourier Transform (Part 2)

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[illegible]

Properties of Fourier Transform (Part 3)

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Fourier Transform **PART**

Properties of Fourier



$\angle$

$$\int_{-\infty}^{\infty} |x(t)| dt < \infty$$

$$\int_{-\infty}^{\infty} A_0 dt = A_0 \int_{-\infty}^{\infty} dt$$

$$= A_0 (t)_{-\infty}^{\infty}$$

$$= A_0 [\infty - (-\infty)]$$

$$= A_0 (\infty + \infty)$$

$$= A_0 \times \infty$$

$$= \infty$$

having the Fourier transform of signal a
naught which

$$x(t) \Rightarrow X(\omega) \text{ or } X(\omega)$$

$$X(\omega) = \int_{-\infty}^{\infty} x(t) \cdot e^{-j\omega t} dt$$

$$= \int_{-\infty}^{\infty} A_0 \cdot e^{-j\omega t} dt$$

$$= A_0 \int_{-\infty}^{\infty} e^{-j\omega t} dt$$

$$= A_0 \left(\frac{e^{-j\omega t}}{-j\omega} \right)_{-\infty}^{\infty}$$

$$= \frac{A_0}{-j\omega} (e^{-j\omega \infty} - e^{-j\omega (-\infty)})$$

$$= \frac{A_0}{\omega} \left(\frac{e^{j\omega \infty} - e^{j\omega (-\infty)}}{j} \right)$$

$$X(\omega) = A_0 \cdot 2 \sin(\omega)$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) \cdot e^{j\omega t} d\omega$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} A_0 \delta(\omega) \cdot e^{j\omega t} d\omega$$

$$= \frac{A_0}{2\pi} \int_{-\infty}^{\infty} \delta(\omega - 0) \cdot e^{j\omega t} d\omega$$

$$= \frac{A_0}{2\pi} \int_{-\infty}^{\infty} \delta(\omega) d\omega$$

1 = Area

$$x(t) = \frac{A_0}{2\pi}$$

$$\frac{A_0}{2\pi} \Rightarrow A_0 \delta(\omega)$$

$$A_0 \Rightarrow 2\pi A_0 \delta(\omega)$$



Properties of Fourier Transform (Part 10)

16. Duality Property:

$$\underline{x(t)} \xRightarrow{\omega = t} \underline{x(\omega)} \xRightarrow{t = -\omega} 2\pi \underline{x(-\omega)}$$

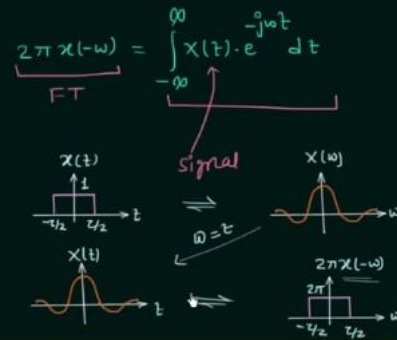
proof :- $x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} x(\omega) \cdot e^{j\omega t} d\omega$

$$t \rightarrow -t$$

$$2\pi x(-t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} x(\omega) \cdot e^{-j\omega t} d\omega$$

$$2\pi x(-t) = \int_{-\infty}^{\infty} x(\omega) \cdot e^{-j\omega t} d\omega$$

$$t \rightarrow \omega \quad \omega \rightarrow t$$



Duality Property of Fourier Transform

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Fourier Transform (DC Value)

$$x(t) \Rightarrow x(\omega) = A_0 \delta(\omega)$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} x(\omega) \cdot e^{j\omega t} d\omega$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} A_0 \delta(\omega) \cdot e^{j\omega t} d\omega$$

$$= \frac{A_0}{2\pi} \int_{-\infty}^{\infty} \delta(\omega - 0) \cdot e^{j\omega t} d\omega$$

$$x(t) = \frac{A_0}{2\pi}$$

$$\frac{A_0}{2\pi} \Rightarrow A_0 \delta(\omega)$$

$$A_0 \Rightarrow 2\pi A_0 \delta(\omega)$$

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Fourier Transform of Basic Signals (DC

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